

Answer Key

Grade 6 Mathematics • Unit 5: Area, Volume and Surface Area • Answers for every item on Sheets 1 and 2 of each lesson packet. Constructed responses show a model answer; accept equivalent reasoning.

LESSON 5-1

Area of Parallelograms • Area of Rhombuses

Sheet 1 • Area of Parallelograms

- $11 \times 6 = 66 \text{ cm}^2$ (the 8 cm side is not used)
- a. 84 ft^2 b. 81 m^2 c. 80 in^2 d. $b = 63 \div 7 = 9 \text{ m}$
- One space: $9 \times 18 = 162 \text{ ft}^2$; 10 spaces: $1,620 \text{ ft}^2$
- $h = 108 \div 12 = 9 \text{ ft}$ (check: $12 \times 9 = 108$)
- She multiplied the base by the slanted side (10×7). The height must be perpendicular to the base: $A = 10 \times 5 = 50 \text{ cm}^2$.

Sheet 2 • Area of Rhombuses

- $(16 \times 10) \div 2 = 80 \text{ cm}^2$
- a. 70 m^2 b. 63 ft^2
- $2 \times 60 = 120$; $120 \div 10 = 12 \text{ ft}$
- A: $(12 \times 8) \div 2 = 48 \text{ ft}^2$; B: $8 \times h = 48$, so $h = 6 \text{ ft}$
- He multiplied the diagonals but did not divide by 2. $A = (18 \times 8) \div 2 = 72 \text{ in}^2$.

TEST PRACTICE

- C
- $C - (20 \times 9) \div 2 = 90 \text{ in}^2$

LESSON 5-2

Area of Triangles • Missing Measures in Triangles

Sheet 1 • Area of Triangles

- $\frac{1}{2} \times 12 \times 7 = 42 \text{ cm}^2$
- $\frac{1}{2} \times 9 \times 8 = 36 \text{ in}^2$
- a. $\frac{1}{2} \times 10 \times 9 = 45 \text{ cm}^2$ b. $\frac{1}{2} \times 5 \times 11 = 27.5 \text{ ft}^2$
- Area = $\frac{1}{2} \times 15 \times 6 = 45 \text{ ft}^2$; $45 \div 9 = 5$ bags
- He forgot to take half (40 m^2 is the parallelogram's area). Correct area: $\frac{1}{2} \times 8 \times 5 = 20 \text{ m}^2$.

Sheet 2 • Missing Measures in Triangles

- A: $h = 2 \times 28 \div 8 = 7 \text{ in}$; B: $b = 2 \times 54 \div 9 = 12 \text{ ft}$
- $2 \times 48 = 96$; $96 \div 16 = 6 \text{ m}$
- a. $2 \times 40 \div 10 = 8 \text{ m}$ b. $2 \times 27 \div 6 = 9 \text{ in}$
- $2 \times 35 = 70$; $70 \div 10 = 7 \text{ ft}$
- Neither. $P = \frac{1}{2} \times 12 \times 5 = 30 \text{ cm}^2$ and $Q = \frac{1}{2} \times 6 \times 10 = 30 \text{ cm}^2$, so the areas are equal.

TEST PRACTICE

- C
- B

LESSON 5-3

Trapezoids by Decomposing • The Trapezoid Area Formula

Sheet 1 • Trapezoids by Decomposing

- a. $6 \times 4 = 24 \text{ cm}^2$ b. triangle base $(10 - 6) \div 2 = 2 \text{ cm}$; $\frac{1}{2} \times 2 \times 4 = 4 \text{ cm}^2$ Area: $24 + 4 + 4 = 32 \text{ cm}^2$
- Rectangle $7 \times 6 = 42$; triangle base $12 - 7 = 5$, $\frac{1}{2} \times 5 \times 6 = 15$; total 57 in^2
- Rectangle $3 \times 4 = 12$; triangle base $8 - 3 = 5$, $\frac{1}{2} \times 5 \times 4 = 10$; total 22 ft^2
- Rectangle $5 \times 3 = 15$; each triangle base $(11 - 5) \div 2 = 3$, $\frac{1}{2} \times 3 \times 3 = 4.5$; total $15 + 4.5 + 4.5 = 24 \text{ ft}^2$
- The two triangle bases together equal $15 - 9 = 6 \text{ cm}$, so each is 3 cm . Rectangle $9 \times 4 = 36$; triangles $2 \times (\frac{1}{2} \times 3 \times 4) = 12$; area = 48 cm^2 .

Sheet 2 • The Trapezoid Area Formula

- $\frac{1}{2} \times (8 + 14) \times 5 = 55 \text{ m}^2$
- a. $\frac{1}{2} \times 14 \times 6 = 42 \text{ ft}^2$ b. $\frac{1}{2} \times 14 \times 8 = 56 \text{ in}^2$ c. $\frac{1}{2} \times 18 \times 3 = 27 \text{ m}^2$ d. $\frac{1}{2} \times 10 \times 9 = 45 \text{ cm}^2$
- $\frac{1}{2} \times (16 + 24) \times 10 = 200 \text{ ft}^2$; $200 \div 25 = 8$ bags
- $2 \times 60 = 120$; $120 \div 6 = 20 = 8 + b$, so $b = 12 \text{ in}$

TEST PRACTICE

- B
- C

LESSON 5-4

Area of Regular Polygons • Composite Figures

Sheet 1 • Area of Regular Polygons

- One triangle $\frac{1}{2} \times 10 \times 12.1 = 60.5$; $8 \times 60.5 \approx 484 \text{ m}^2$
- Hexagon: $P = 24 \text{ cm}$, $A \approx \frac{1}{2} \times 24 \times 3.5 = 42 \text{ cm}^2$; Pentagon: $P = 50 \text{ ft}$, $A \approx \frac{1}{2} \times 50 \times 6.9 = 172.5 \text{ ft}^2$

Sheet 2 • Composite Figures

- Bottom $10 \times 4 = 40$; top $4 \times 4 = 16$; total 56 m^2
- a. subtract (wall area – window area) b. add (room area + closet area)
- Top bar $10 \times 2 = 20$; stem $4 \times 4 = 16$; total 36 in^2

3. a. $\approx \frac{1}{2} \times 30 \times 4.1 = 61.5 \text{ cm}^2$ b. $\approx \frac{1}{2} \times 60 \times 8.7 = 261 \text{ ft}^2$
4. Area $\approx 6 \times (\frac{1}{2} \times 6 \times 5.2) = 93.6 \text{ ft}^2$; cost $\approx 93.6 \times \$3 = \280.80
5. He found only one of the 5 triangles. Area $\approx 5 \times 22 = 110 \text{ cm}^2$.

4. Rectangle $12 \times 8 = 96$; notch base $12 - 3 - 3 = 6$, $\frac{1}{2} \times 6 \times 4 = 12$; $96 - 12 = 84 \text{ cm}^2$
5. $240 - 25 = 215 \text{ ft}^2$

TEST PRACTICE

6. A
7. C

LESSON 5-5

Volume of Rectangular Prisms · Missing Fractional Edges

Sheet 1 · Volume of Rectangular Prisms

1. $4 \times 3 \times 2 = 24 \text{ cm}^3$
2. a. 80 in^3 b. 400 cm^3 c. 216 ft^3 d. 315 m^3 e. 180 m^3 f. 288 in^3
3. $V = 6 \times 4 \times 1 = 24 \text{ ft}^3$; $24 \div 2 = 12$ bags
4. $B = 20 \times 10 = 200 \text{ in}^2$; $h = 3,000 \div 200 = 15 \text{ in}$
5. No. $A = 6 \times 4 \times 2 = 48 \text{ in}^3$ and $B = 3 \times 4 \times 4 = 48 \text{ in}^3$. They hold the same amount.

Sheet 2 · Missing Fractional Edges

1. $4 \times \frac{3}{2} \times \frac{5}{2} = 60/4 = 15 \text{ in}^3$
2. a. 24 ft^3 b. 28 m^3 c. $\frac{5}{2} \times 12 = 30 \text{ cm}^3$ d. $\frac{9}{4} \times 4 = 9 \text{ in}^3$ e. $4 \times \frac{1}{2} \times 5 = 10 \text{ ft}^3$ f. $6 \times \frac{3}{2} \times 3 = 27 \text{ cm}^3$
3. $3 \times 1 \frac{1}{2} = 4 \frac{1}{2} \text{ ft}^2$; $9 \div 4 \frac{1}{2} = 2 \text{ ft}$
4. a. $4 \times 2 \times 3 = 24$ cubes b. $24 \times \frac{1}{8} = 3 \text{ in}^3$ (check: $2 \times 1 \times 1 \frac{1}{2} = 3$)

TEST PRACTICE

5. D
6. A

LESSON 5-6

Drawing Nets of Solids · Reading Nets and Solids

Sheet 1 · Drawing Nets of Solids

1. Net of 6 rectangles: two $6 \text{ in} \times 2 \text{ in}$, two $6 \text{ in} \times 4 \text{ in}$, two $2 \text{ in} \times 4 \text{ in}$, joined so they fold with no overlaps.
2. a. 6 b. 4 c. 3 d. 5
3. a. $8 \times 8 = 64 \text{ cm}^2$ b. $\frac{1}{2} \times 8 \times 6 = 24 \text{ cm}^2$ All 4 triangles: 96 cm^2
4. two $8 \text{ in} \times 3 \text{ in}$, two $8 \text{ in} \times 12 \text{ in}$, two $3 \text{ in} \times 12 \text{ in}$
5. No. The row wraps around 4 sides, and the last 2 squares overlap the first 2. Nothing covers the top or bottom. A cube net needs squares attached above and below the row.

Sheet 2 · Reading Nets and Solids

1. triangular prism; 5 faces
2. a. rectangular pyramid b. hexagonal prism c. triangular prism d. pentagonal pyramid e. cube f. pentagonal prism
3. a. $6 \times 6 = 36 \text{ in}^2$ b. $4 \times (\frac{1}{2} \times 6 \times 5) = 60 \text{ in}^2$
4. Rectangular prism (square prism), $4 \text{ cm} \times 4 \text{ cm} \times 9 \text{ cm}$
5. C – 3 edges on each triangle + 3 joining edges = 9
6. B – 1 hexagon + 6 triangles

TEST PRACTICE

LESSON 5-7

Surface Area of Rectangular Prisms · Surface Area of Triangular Prisms

Sheet 1 · Surface Area of Rectangular Prisms

1. $2(10 \times 4) + 2(10 \times 5) + 2(4 \times 5) = 80 + 100 + 40 = 220 \text{ cm}^2$
2. a. 54 cm^2 b. 600 ft^2 c. 28 m^2 d. 90 in^2 e. $6 \times 25 = 150 \text{ in}^2$ f. $2(18) + 2(12) + 2(6) = 72 \text{ cm}^2$
3. $SA = 2(96) + 2(48) + 2(32) = 352 \text{ in}^2$. Yes, $400 > 352$, with 48 in^2 to spare.
4. Top $4 \times 2 = 8$; front and back $2 \times (4 \times 3) = 24$; sides $2 \times (2 \times 3) = 12$; $8 + 24 + 12 = 44 \text{ ft}^2$
5. He found the volume (8 cm^3). Surface area adds the 6 faces: $6 \times (2 \times 2) = 24 \text{ cm}^2$.

Sheet 2 · Surface Area of Triangular Prisms

1. Triangles $2 \times (\frac{1}{2} \times 6 \times 4) = 24$; rectangles $6 \times 10 + 5 \times 10 + 5 \times 10 = 160$; $SA = 184 \text{ m}^2$
2. a. $2 \times (\frac{1}{2} \times 8 \times 3) = 24 \text{ ft}^2$ b. 45 ft^2
3. Ends 24 ; floor $8 \times 9 = 72$; sides $2 \times (5 \times 9) = 90$; total 186 ft^2
4. Ends $2 \times (\frac{1}{2} \times 5 \times 12) = 60$; rectangles $5 \times 4 + 12 \times 4 + 13 \times 4 = 120$; $SA = 180 \text{ in}^2$
5. He used the triangle's height (4 m) as a side. The rectangles use the sides of the triangle: 6×10 , 5×10 , and 5×10 .

TEST PRACTICE

6. C
7. $B - 6 \times 8 + 8 \times 8 + 10 \times 8 = 192 \text{ cm}^2$

LESSON 5-8

Surface Area of Pyramids · Lateral Area of Pyramids

Sheet 1 · Surface Area of Pyramids

1. $36 + 4 \times (\frac{1}{2} \times 6 \times 9) = 36 + 108 = 144 \text{ m}^2$

Sheet 2 · Lateral Area of Pyramids

1. $4 \times (\frac{1}{2} \times 8 \times 6) = 96 \text{ ft}^2$

2. a. 144 cm^2 b. $\frac{1}{2} \times 12 \times 10 = 60 \text{ cm}^2$ SA = $144 + 240 = 384 \text{ cm}^2$
3. a. $16 + 4 \times (\frac{1}{2} \times 4 \times 5) = 56 \text{ in}^2$ b. $100 + 4 \times (\frac{1}{2} \times 10 \times 12) = 340 \text{ cm}^2$
4. $81 + 4 \times (\frac{1}{2} \times 9 \times 11) = 81 + 198 = 279 \text{ in}^2$
5. He used the pyramid's height (4 ft) instead of the slant height (5 ft). SA = $36 + 4 \times (\frac{1}{2} \times 6 \times 5) = 36 + 60 = 96 \text{ ft}^2$.
2. a. 136 in^2 b. 72 cm^2 c. $4 \times (\frac{1}{2} \times 6 \times 10) = 120 \text{ m}^2$ d. $4 \times (\frac{1}{2} \times 3 \times 4) = 24 \text{ ft}^2$
3. Lateral area $4 \times (\frac{1}{2} \times 14 \times 10) = 280 \text{ ft}^2$; $280 \div 35 = 8$ bundles
4. One face: $160 \div 4 = 40 \text{ in}^2$; $\frac{1}{2} \times 8 \times \text{slant height} = 40$, so slant height = 10 in
5. No. $85 \text{ ft}^2 = 25 + 60$ includes the floor. With no floor, only the lateral area is needed: $4 \times (\frac{1}{2} \times 5 \times 6) = 60 \text{ ft}^2$.

TEST PRACTICE

6. B
7. C